

Gale–Shapley Algorithm: Men-Proposing Stable Matching

Algorithm type: Greedy / iterative algorithm for the Stable Marriage (Stable Matching) Problem. The men-proposing version is a deferred-acceptance algorithm: proposals are made according to each man's highest-ranked untried choice, while each woman keeps her preferred proposal so far.

Given Preferences

Men	Preferences
m1	w3 > w1 > w4 > w2 > w5
m2	w1 > w3 > w2 > w5 > w4
m3	w2 > w5 > w1 > w3 > w4
m4	w2 > w3 > w5 > w4 > w1
m5	w4 > w2 > w3 > w1 > w5

Women	Preferences
w1	m3 > m2 > m5 > m1 > m4
w2	m5 > m1 > m4 > m3 > m2
w3	m4 > m2 > m1 > m5 > m3
w4	m2 > m5 > m3 > m4 > m1
w5	m1 > m4 > m3 > m2 > m5

Gale–Shapley Procedure

- Initially, everyone is unmatched.
- An unmatched man proposes to the highest-ranked woman he has not yet proposed to.
- If the woman is unmatched, she accepts provisionally.
- If she is matched, she keeps whichever man she prefers and rejects the other.
- A rejected man becomes free and proposes to his next choice.
- Continue until every man is matched. The final matching is stable.

Step-by-Step Solution

Step	Man	Woman	Result
1	m1	w3	Accepted
2	m2	w1	Accepted
3	m3	w2	Accepted
4	m4	w2	Accepted (replaces m3)
5	m5	w4	Accepted
6	m3	w5	Accepted

Final Stable Matching

(m1, w3), (m2, w1), (m3, w5), (m4, w2), (m5, w4)

Therefore: m1–w3, m2–w1, m3–w5, m4–w2, and m5–w4.

Why is it stable?

Because Gale–Shapley terminates only after every man is matched, and the deferred-acceptance process guarantees that no man and woman can form a blocking pair. A woman has rejected every man she prefers less than her final partner, so a man cannot be able to obtain a preferred woman who would also prefer him over her final partner.